P 4.2 Analytical Procedures

Analytical procedures of the excipients used in the manufacturing of Solifenacin Succinate Tablets BP 5 mg are compendial and have been analyzed as per the test methods mentioned in the respective monographs.

Sr. No.	Ingredients	Specifications
1	Lactose Monohydrate	BP
2	Maize Starch	BP
3	Hypromellose	BP
4	Magnesium Stearate	BP
5	Ready coat 36FO193220	IH
6	Purified Water	BP

BP: British Pharmacopoeia

IH: In House

The analytical procedures used in Solifenacin Succinate Tablets BP 5 mg are enlisted below:

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			Supersedes	Nil
Quality Control Department	Raw Material Standard Test Procedure		Effective Date	07/12/2020
	Item Name	Solifenacin Succinate BP		
	Reference	BP	Market	Export

1. **Description:**
   Instruments: Analytical Balance.
   Reagents: NA
   Procedure: Take about 1 g of test sample in a petri dish and note down the observation by inspecting visually.

2. **Solubility:**
   Instruments : Analytical Balance
   Reagents : ethanol,heptane
   Procedure:
   Water: Take 1 g of sample into test tube Add 1 to 10 ml of water and check for solubility
   Ethanol: Take 1 g of sample into test tube Add 10 to 30 ml of ethanol and check for solubility.
   Heptane: Take 10 mg of sample into test tube Add more than 100 ml of heptane and check for solubility.

3. **Identification**
   A. By IR:
   Instruments : Analytical Balance, IR
   Reagents : NA
   Procedure: Take about 2 mg of solifenacin succinate WS/RS place it on ATR assembly crystal and record the spectrum of solifenacin succinate WS/RS.
   Take about 5 mg of test sample placed it onto the ATR crystal and record the spectrum of test sample.
   Compare the spectrum of test sample with that obtained with solifenacin succinate WS/RS or with the reference spectrum of solifenacin succinate.

   B. By Isomeric Purity:
   Instruments : Analytical Balance, HPLC
   Reagents : NA
   Procedure: Observe result obtained in the test of isomeric purity.

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4. **Isomeric Purity:**

Instruments : Analytical Balance, HPLC

Reagents : diethylamine, anhydrous ethanol, heptane

Procedure :

Mobile Phase: diethylamine, anhydrous ethanol, heptane (0.1:200:800 V/V/V).

Test solution: Dissolve 50 mg of the substance to be examined in the mobile phase and dilute to 20.0 mL with the mobile phase.

Standard Solution A: Dissolve the contents of a vial of solifenacin for system suitability CRS (containing impurities F, G and H) in 1.0 mL of the mobile phase.

Standard Solution B: Dilute 1.0 mL of the test solution to 100.0 mL with the mobile phase. Dilute 1.0 mL of this solution to 10.0 mL with the mobile phase.

Chromatographic conditions:

Column	size: l = 0.25 m, Ø = 4.6mm;stationary phase: : amylose derivative of silica gel for chiral separation R (5 µm);
Column temperature	35 °C.
Injection volume	10µl
Detection	Spectrophotometer at 220 nm
Run time	1.5 times the retention time of solifenacin
Flow rate	0.8 ml per min

Identification of impurities: use the chromatogram supplied with solifenacin for system suitability CRS and the chromatogram obtained with standard solution (A) to identify the peaks due to impurities F, G and H.

Solution	No. of injections
Blank	1
Standard solution A	1

Standard solution B	5
Test solution	1
Bracketing Standard solution B	1
Bracketing Standard solution A	1

Retention time for solifenacin is about 17 minutes.

Relative Retention Time:

Impurity F	about 0.7
Impurity H	about 0.76
Impurity G	about 0.84

System suitability: Standard solution (A): – resolution: minimum 1.5 between the peaks due to impurities F and H;

– peak-to-valley ratio: minimum 10, where Hp = height above the baseline of the peak due to impurity G and Hv = height above the baseline of the lowest point of the curve separating this peak from the peak due to impurity H

Calculation:

$$\text{Impurity F} = \frac{AF}{As(B)} \times \frac{Wt}{20} \times \frac{1}{100} \times \frac{1}{10} \times \frac{20}{Wt} \times 100$$

$$\text{Impurity H} = \frac{AH}{As(B)} \times \frac{Wt}{20} \times \frac{1}{100} \times \frac{1}{10} \times \frac{20}{Wt} \times 100$$

$$\text{Impurity G} = \frac{AG}{As(B)} \times \frac{Wt}{20} \times \frac{1}{100} \times \frac{1}{10} \times \frac{20}{Wt} \times 100$$

AF = Peak area obtained in test solution for impurity F.
AH = Peak area obtained in test solution for impurity H.
AG = Peak area obtained in test solution for impurity G.
As(B) = Peak area obtained in the standard solution B.
Wt = Weight of the sample in test solution (mg).

5. **Impurity E by TLC :**

Instruments : Analytical Balance,

Reagents : methanol, ammonia, anhydrous ethanol, toluene

Procedure: Determine by thin-layer chromatography, coating the plate with silica gel G

Mobile phase : concentrated ammonia , anhydrous ethanol , toluene (6:31:63 V/V/V)Apply separately to the plate 10 µl of each of the following solutions

Test solution (A): Dissolve 500 mg of the substance to be examined in methanol and dilute to 5.0 mL with the same solvent.

Standard solution (B): Dissolve 5.0 mg of 3-quinuclidinol (impurity E) in methanol and dilute to 50.0 mL with the same solvent.

Development: 2/3 of the plate.

After removal of the plate, allow it to dry in oven at 100 to 105°C for 1 hour andexpose to iodine vapour for at least 2 hours. The principal spots in the chromatogram obtained with solution (A) correspond to the principal spots in the chromatogram obtained with solution (B).

6. **Related substances:**

Instruments : Analytical Balance, HPLC

Reagents : ammonium phosphate, diethylamine, ammonia,

Procedure:

Solvent mixture: Dissolve 600mg of ammonium phosphate in 1 L of water , add 1.0 mL of diethylamine and adjust to pH 11±0.05 with concentrated ammonia.

Test solution: Dissolve 100 mg of the substance to be examined in mobile phase A and dilute to 100.0 mL with mobile phase A.

Standard solution (A): Dissolve 5 mg of 1-phenyl-1,2,3,4- tetrahydroisoquinoline (impurity A) in the test solution and dilute to 25.0 mL with the test solution.

Standard solution (B): Dilute 1.0 mL of the test solution to 100.0 mL with mobile phase A. Dilute 1.0 mL of this solution to 10.0 mL with mobile phase A

Mobile phase: – Mobile phase A: acetonitrile R1, solvent mixture (45:55 V/V);
– Mobile phase B: acetonitrile R1;

Time (min)	Mobile phase A (per cent V/V)	Mobile phase B (per cent V/V)
0 - 0.5	90	10

0.5 - 10.5	90 → 68	10 → 32
10.5 - 12.5	68	32

Chromatographic conditions:

Column	size: l = 0.10 m, Ø = 2.1mm;end-capped ethylene-bridged polar-embedded octadecylsilyl silica gel for chromatography (hybrid material) R (1.7 µm);
Column temperature	40 °C.
Injection volume	2.5 µl
Detection	Spectrophotometer at 215 nm.
Flow rate	0.4 ml per min

Solution	No. ofinjections
Blank	1
Standard solution A	1
Standard solution B	5
Testsolution	1
Bracketing StandardsolutionB	1
Bracketing Standard solution A	1

Identification of impurities: use the chromatogram obtained with reference solution (a) to identify the peak due to impurity A.

Relative retention time

solifenacin	about 2 min
succinate	about 0.2 min
impurity A	about 0.7 min

System suitability: Standard solution (A): – resolution: minimum 4.0 between the peaks due to impurity A and solifenacin.

Calculation:

Unknown Impurity= $\frac{AU}{As(B)} \times \frac{Wt}{100} \times \frac{1}{100} \times \frac{1}{10} \times \frac{100}{Wt} \times 100$

$$\text{TotalImpurity} = \frac{AT}{As(B)} \times \frac{Wt}{100} \times \frac{1}{100} \times \frac{1}{10} \times \frac{100}{Wt} \times 100$$

AU = Peak area obtained in test solution for Unknown impurity.
AH = Total Peak area due to impurities obtained in test solution.
As(B) = Peak area obtained in the standard solution B.
Wt = Weight of the sample in test solution (mg).

7. **Loss on drying:**

Instrument : Analytical Balance, vacuum oven,
Reagents : NA
Procedure: Weigh a glass stoppered weighing bottle (Wl) with as it is previously dried for 30 min at 105±2°C and cooled in a desiccators. Transfer about 1.0 g of sample in bottle, cover it and accurately weigh (W2), remove the stopper and dry the sample in an oven at 105±2°C for 3 hours. Open the chamber Close the bottle promptly and allow the sample in a desiccator to come room temperature, weigh the bottle and record the weight (W3).

$$\text{Loss on drying (\%w/w)} = \frac{(W2-W3)}{(W2-W1)} \times 100$$

Where,
W_1= Weight of empty bottle
W_2 = Weight of bottle with sample
W_3 = Weight of bottle with sample after drying

8. **Sulfated ash:**

Instruments : Analytical Balance, kf auto titrator
Reagents : Sulphuric acid.
Procedure: Heat a silica or platinum crucible to redness for 10 minutes, allow to cool in a desiccator and weigh.(W1) Transfer 1.0 g of sample to the crucible weigh the crucible and thecontents accurately.(W2) Ignite, gently at first, until the sample is thoroughly charred. Cool, moisten the residue with 1 ml ofsulphuric acid, heat gently until the white fumes are no longer evolved and ignite at 600°C ± 50°C until all black particles have disappeared. Conduct the ignition in a place protected from air currents. Allow the crucible to cool, add a few drops of sulphuric acid and heat. Ignite as before, allow to cool and weigh (W4). Repeat the

operation until two successive weighing do not differ by more than 0.5 mg to get constant weight (W5).

$$\text{Sulphated Ash (\%w/w)} = \frac{(W5-W1)}{W3} \times 100$$

Where,

W1 = weight of empty crucible
W2 = weight of crucible with sample
W3 = (W2-W1)
W4 = weight after ignition
W5 = constant weight

9. Assay:

Instruments : Analytical Balance, kf auto titrator
Reagents : Anhydrous acetic acid, Perchloric acid
Procedure: Dissolve 400 mg in 50 mL of anhydrous acetic acid. Titrate with 0.1 M perchloric acid, determining the end-point potentiometrically.
1 mL of 0.1 M perchloric acid is equivalent to 48.06 mg of $C_{27}H_{32}N_2O_6$.

$$= \frac{\text{End point (Blank – Test)} \times \text{Factor} \times \text{Actual Molarity} \times 100}{\text{Weight of Sample} \times \text{Stated Molarity}}$$

Remark: As per new format SOP No. SOP/QCD/086

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			Supersedes	Nil
Quality Control Department	Raw Material Standard Test Procedure		Effective Date	10/12/2020
	Item Name	Lactose Monohydrate BP		
	Reference	BP	Market	Export

1. **Description:**
   Instruments : Analytical Balance
   Reagents : NA
   Procedure: Take 2g of Test sample in a petri dish and inspect visually.

2. **Solubility:**
   Instruments : Analytical Balance
   Reagents : Ethanol
   Procedure:
   Water: Take 1 g of sample in test tube add 10 to 30 ml of water and check for solubility.
   Ethanol (96%): Take 10 mg of sample in test tube add more than 100 ml of water and check for solubility.

3. **Identification:**
   A. By IR
   Instruments : Analytical Balance, IR spectrophotometer
   Reagents : NA
   Procedure: Compare the spectra of test sample with anhydrous lactose CRS.

   B. By TLC
   Instruments : Analytical Balance
   Reagents : methanol, fructose, glacial acetic acid, ethylene chloride, thymol
   Procedure:
   Solvent mixture: water R, methanol R (40:60 V/V).
   Test solution: Dissolve 10 mg of the substance to be examined in the solvent mixture and dilute to 20 mL with the solvent mixture.

   Reference solution (a): Dissolve 10 mg of anhydrous lactose CRS in the solvent mixture and dilute to 20 mL with the solvent mixture.

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Reference solution (b): Dissolve 10 mg of fructose R, 10 mg of glucose R, 10 mg of lactose monohydrate R and 10 mg of sucrose R in the solvent mixture and dilute to 20 mL with the solvent mixture.

Plate: TLC silica gel plate R.

Mobile phase: water R, methanol R, glacial acetic acid R, ethylene chloride R (10:15:25:50 V/V/V/V); measure the volumes accurately, as a slight excess of water produces cloudiness.

Application: 2 µL; thoroughly dry the points of application.

Development A: Over 3/4 of the plate.

Drying A: In a current of warm air.

Development B: Immediately, over 3/4 of the plate, after renewing the mobile phase.

Drying B: In a current of warm air.

Detection: Spray with a solution of 0.5 g of thymol R in a mixture of 5 mL of sulfuric acid R and 95 mL of ethanol (96 per cent) R; heat at 130 °C for 10 min.

System suitability: reference solution (b):
— the chromatogram shows 4 clearly separated spots.

C. By Reaction
Instruments : Analytical Balance
Reagents : ammonia,
Procedure: Dissolve 0.25 g in 5 mL of water R. Add 5 mL of ammonia R and heat in a water-bath at 80 °C for 10 min. A red colour develops.

D. By Water
Instruments : Analytical Balance
Reagents : NA
Procedure: See the test

4. **Appearance of solution:**

Instruments : Analytical Balance

Reagents : hydrochloric acid, sodium thiosulfate, copper sulfate pentahydrate, sulfuric acid

Procedure:

Solution S: Dissolve 1.0 g in boiling water R, allow to cool and dilute to 10.0 mL with water.

For Clarity of the Solution:

Using identical test tubes of colourless, transparent, neutral glass with a flat base and an internal diameter of 15 – 25 mm, compare the liquid to be examined with a reference suspension freshly prepared as described below. Ensure that the depths of the layers in the 2 test-tubes are the same (about 40 mm).

Standard of Opalescence: Dilute 15.0 ml of the primary opalescent suspension to 1000.0 ml with water. This suspension is freshly prepared and may be stored for up to 24h.

Primary opalescent suspension (formazin suspension): Dissolve 2.5 g of hexamethylenetetramine in 100 ml of ground glass stoppered flask, add 25.0 ml water. Add 25.0 ml of hydrazine sulfate solution. Mix and allow to stand for 24h. This suspension is stable for 2 months, provided it is stored in a glass container free from surface defects. The suspension must not adhere to the glass and must be mixed thoroughly before use.

Hydrazine Sulfate Solution: Dissolve 1.0 g of hydrazine sulfate in water and dilute to 100.0 ml with same solvent. Allow to stand for 4-6h.

Reference Suspensions:

Prepare the reference suspensions given below and compare with test solution for clarity of solution.

	I	II	III	IV
Standard of opalescence	5.0 ml	10.0 ml	30.0 ml	50.0 ml
Water	95.0 ml	90.0 ml	70.0 ml	50.0 ml

For Colour of the Solution:

Preparation of Reference Solution:

Yellow solution: Dissolve 46 g of ferric chloride in about 900 mL of a mixture of 25 mL of hydrochloric acid and 975 mL of water and dilute to 1000.0 mL with the same mixture. Titrate and adjust the solution to contain 45.0 mg of FeCl3,6H2O per milliliter by adding the same acidic mixture. Protect the solution from light.

Titration: Place in a 250 mL conical flask fitted with a ground-glass stopper, 10.0 mL of the solution, 15 mL of water, 5 mL of hydrochloric acid and 4 g of potassium iodide, close the flask, allow to stand in the dark for 15 min and add 100 mL of water. Titrate the liberated iodine with 0.1 M sodium thiosulfate, using 0.5 mL of starch solution, added towards the end of the titration, as indicator.

1 mL of 0.1 M sodium thiosulfate is equivalent to 27.03 mg of FeCl3,6H2O.

Red solution: Dissolve 60 g of cobalt chloride in about 900 mL of a mixture of 25 mL of hydrochloric acid and 975 mL of water and dilute to 1000.0 mL with the same mixture. Titrate and adjust the solution to contain 59.5 mg of CoCl2,6H2O per milliliter by adding the same acidic mixture.

Titration: Place in a 250 mL conical flask fitted with a ground-glass stopper, 5.0 mL of the solution, 5 mL of dilute hydrogen peroxide solution and 10 mL of a 300 g/L solution of sodium hydroxide. Boil gently for 10 min, allow to cool and add 60 mL of dilute sulfuric acid and 2 g of potassium iodide. Close the flask and dissolve the precipitate by shaking gently. Titrate the liberated iodine with 0.1 M sodium thiosulfate, using 0.5 mL of starch solution, added towards the end of the titration, as indicator. The end-point is reached when the solution turns pink.

1 mL of 0.1 M sodium thiosulfate is equivalent to 23.79 mg of CoCl2.6H2O.

Blue solution: Dissolve 63 g of copper sulfate pentahydrate R in about 900 mL of a mixture of 25 mL of hydrochloric acid R and 975 mL of water R and dilute to 1000.0 mL with the same mixture. Titrate and adjust the solution to contain 62.4 mg of CuSO4,5H2O per milliliter by adding the same acidic mixture.

Titration: Place in a 250 mL conical flask fitted with a ground-glass stopper, 10.0 mL of the solution, 50 mL of water R, 12 mL of dilute acetic acid R and 3 g of potassium iodide R. Titrate the liberated iodine with 0.1 M sodium thiosulfate, using 0.5 mL of starch solution R, added towards the end of the titration, as indicator. The end-point is reached when the solution shows a slight pale brown colour.

1 mL of 0.1 M sodium thiosulfate is equivalent to 24.97 mg of CuSO4,5H2O.

Preparation of standard solution BY: Add 2.4 ml of yellow solution, 1.0 ml of red solution, 0.4 ml of blue solution in to 10 ml volumetric flask and make up the volume with 10g/ml HCL.

Preparation of Reference solution BY_7: Add 2.5 ml of standard solution BY into a 100 ml volumetric flask and make up the volume with 10g/ml HCL.

5. **Acidity or alkalinity**

Instruments : Analytical Balance
Reagents : phenolphthalein solution, sodium hydroxide
Procedure:
Dissolve 6.0 g by heating in 25 mL of carbon dioxide-free water R, cool and add 0.3 mL of phenolphthalein solution R1. The solution is colourless. Not more than 0.4 mL of 0.1 M sodium hydroxide is required to change the colour of the indicator to pink or red.

6. **Specific Optical rotation**

Instruments : Analytical Balance
Reagents : ammonia
Procedure: Dissolve 10.0 g in 80 mL of water R with heating at 50 °C. Allow to cool and add 0.2 mL of dilute ammonia R1. Allow to stand for 30 min and dilute to 100.0 mL with water R.

7. **Absorbance**

Instruments : Analytical Balance
Reagents : NA
Procedure:
Test solution(a): Solution S.
Test solution(b): Dilute 1.0 mL of test solution (a) to 10.0 mL with water R.
Spectral range: 400 nm for test solution (a) and 210-300 nm for test solution (b).

8. **Water:**

Instruments : Analytical Balance, KF autotitrator
Reagents : dehydrated methanol, KF reagent
Procedure:

Solvent mixture: mixture of 1 volume of formamide and 2 volumes of methanol.
Add about sufficient volume of Solvent mixture to the titration vessel and titrate to the electrometric end point with the KF reagent. Weigh 0.5 g of sample and transfer quickly to the titration vessel stir for 1 min and titrate again to the electrometric end point using KF reagent.

9. **Sulphated ash:**

Instruments : Analytical Balance, Muffle Furnace
Reagents : Sulphuric acid.
Procedure: Heat a silica or platinum crucible to redness for 10 minutes, allow to cool in a desiccator and weigh. (W1) Transfer 1 g of sample to the crucible weigh the crucible and the contents accurately. (W2) Ignite, gently at first, until the sample is thoroughly charred. Cool, moisten the residue with 1 ml of sulphuric acid, heat gently until the white fumes are no longer evolved and ignite at 800° ± 25° until all black particles have disappeared. Conduct the ignition in a place protected from air currents. Allow the crucible to cool, add a few drops of sulphuric acid and heat. Ignite as before, allow to cool and weigh(W4). Repeat the operation until two successive weighing do not differ by more than 0.5 mg to get constant weight (W5)

$$\text{Sulphated Ash (\%w/w)} = \frac{(W5-W1)}{W3} \times 100$$

Where,
W1 = weight of empty crucible
W2 = weight of crucible with sample
W3 = (W2-W1)
W4 = weight after ignition
W5 = constant weight

10. **Microbial Contamination:** Refer Microbiological SOP/QCM/039

Remark: As per new format SOP No. SOP/QCD/086

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			Supersedes	Nil
Quality Control Department	Raw Material Standard Test Procedure		Effective Date	12/12/2020
	Item Name	Maize starch BP		
	Reference	BP	Market	Export

1. **Description:**
   Instruments :Analytical Balance.
   Reagents :NA
   Procedure: Take Sufficient test sample in a petri dish and inspect visually.

2. **Solubility :**
   Instruments : Analytical Balance.
   Reagents : NA
   Procedure: Practically insoluble in cold water and in ethanol (96 %)

3. **Identification:**
   A. By Microscope Examination:
   Instruments : Microscope.
   Reagents : NA
   Procedure : using a 50 per cent v/v solution of glycerol. It appears as either angular polyhedral granules of irregular sizes with diameters ranging from about 2 µm to about 23 µm or as rounded or spheroidal granules of irregular sizes with diameters ranging from about 25 µm to about 35 µm. The central hilum consists of a distinct cavity or 2- to 5-rayed cleft and there are no concentric striations. Between orthogonally orientated polarising plates or prisms, the starch granules show a distinct black cross intersecting at the hilum.

   B. By Chemical:
   Instruments : Analytical Balance.
   Reagents : NA
   Procedure: Suspend 1 g in 50 ml of water, boil for 1 min and cool. A thin, cloudy mucilage is formed.
   By Chemical :
   Instruments : NA
   Reagents : Iodine solution

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			Supersedes	Nil
Quality Control Department	Raw Material Standard Test Procedure		Effective Date	12/12/2020
	Item Name	Maize starch BP		
	Reference	BP	Market	Export

7. **Sulfur Dioxide :**

Instruments : Analytical Balance.

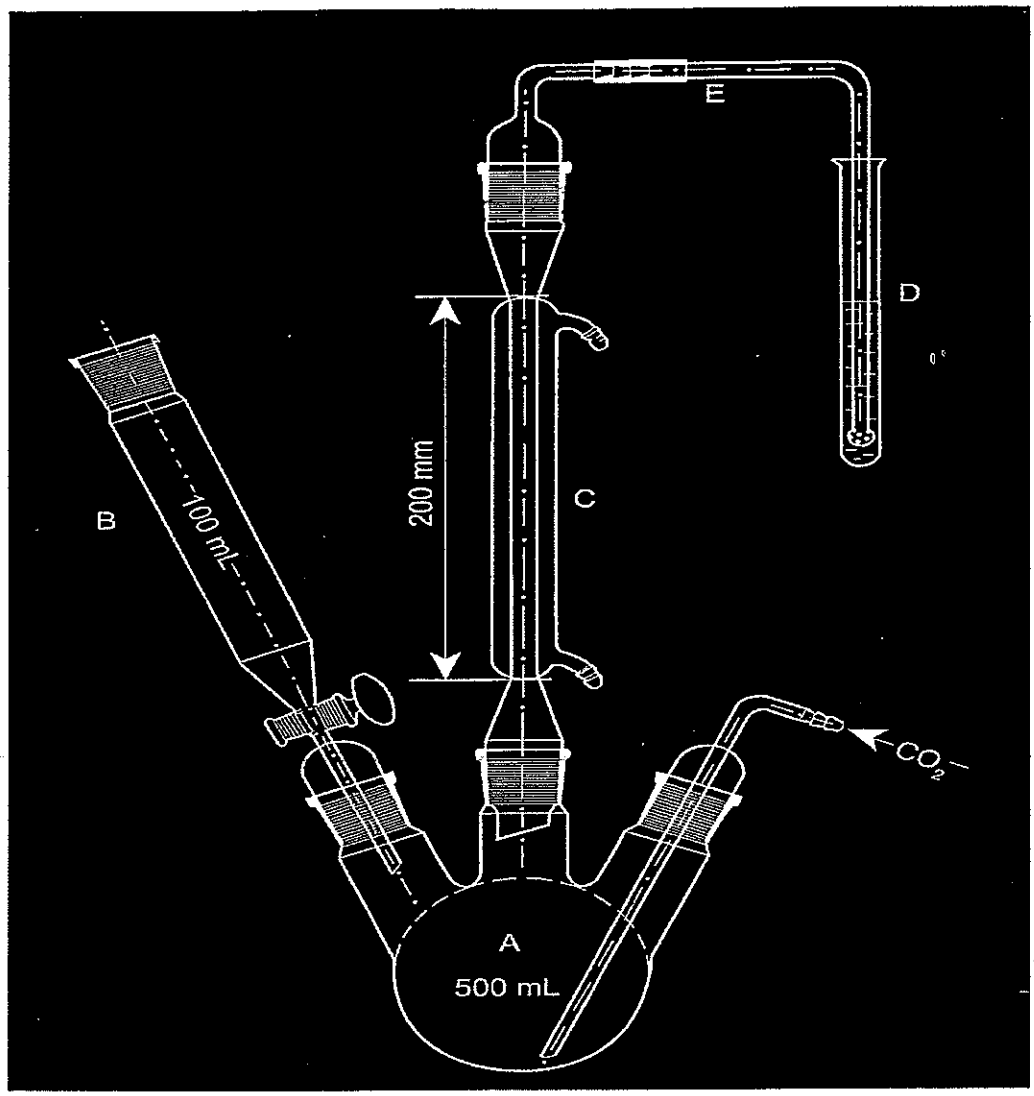
Reagents : Hydrochloric acid, bromophenol blue solution, 0.1M NaoH,

Procedure : Introduce 150 ml of water into the flask(A) and pass carbon dioxide through the whole system for 15 min at a rate of 100 ± 5 ml/min. To 10 ml of dilute hydrogen peroxide solution add 0.15 ml of a 1 g/L solution of bromophenol blue in ethanol (20 per cent v/v). Add 0.1 M sodium hydroxide until a violet-blue colour is obtained, without exceeding the end-point. Place the solution in the test-tube (D). Without interrupting the stream of carbon dioxide, remove the funnel (B) and introduce through the opening into the flask (A) 25.0 g (m g) of the substance to be examined with the aid of 100 ml of water. Replace the funnel. Close the tap of the funnel and add 80 ml of dilute hydrochloric acid to the funnel. Open the tap of the funnel to allow the hydrochloric acid solution to flow into the flask, making sure that no sulfur dioxide escapes into the funnel by closing the tap before the last few millilitres of hydrochloric acid solution drain out. Boil for 1 h. Open the tap of the funnel and stop the flow of carbon dioxide and also the heating and the cooling water. Transfer the contents of the test-tube with the aid of a little water to a 200 ml wide-necked, conical flask. Heat on a water-bath for 15 min and allow to cool. Add 0.1 ml of a 1 g/L solution of bromophenol blue in ethanol (20 per cent v/v) and titrate with 0.1 M sodium hydroxide until the colour changes from yellow to violet-blue (V_1 mL). Carry out a blank titration (V_2 mL).

Refer following image for arrangement of apparatuses

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			Supersedes	Nil
Quality Control Department	Raw Material Standard Test Procedure		Effective Date	12/12/2020
	Item Name	Maize starch BP		
	Reference	BP	Market	Export



32030 x ($V1-V_2$) X n / m

n = molarity of the sodium hydroxide solution used as titrant.

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Procedure : To 1 ml of the mucilage obtained in identification test B add 0.05 ml of iodine solution. An orange-red to dark blue colour is produced, which disappears on heating.

4. **pH:**
   Instruments : Analytical Balance, pH meter
   Reagents : NA
   Procedure : To 5.0 g add 25.0 ml of carbon dioxide-free water. Agitate continuously at a moderate rate for 60 s. Stop the agitation and allow to stand for 15 min.

5. **Foreign matter :**
   Instruments : Microscope
   Reagents : NA
   Procedure : Examined under a microscope using a mixture of equal volumes of glycerol and water, not more than traces of matter other than starch granules are present. No starch grains of any other origin are present.

6. **Oxidizing Substance:**
   Instruments : Analytical balance
   Reagents : Acetic acid, potassium iodide, 0.002M sodium thiosulphate, Starch
   Procedure : Transfer 4.0 g to a glass-stoppered, 125 ml conical flask and add 50.0 ml of water. Insert the stopper and swirl for 5 min. Transfer to a glass-stoppered 50 ml centrifuge tube and centrifuge. Transfer 30.0 ml of the clear supernatant to a glass-stoppered 125 ml conical flask. Add 1 mL of glacial acetic acid and 0.5 g to 1.0 g of potassium iodide. Insert the stopper, swirl, and allow to stand for 25 min to 30 min in the dark. Add 1 ml of starch solution and titrate with 0.002 M sodium thiosulfate until the starch-iodine colour disappears. Carry out a blank determination. Not more than 1.4 mL of 0.002 M sodium thiosulfate is required (0.002 per cent, calculated as H_2O_2).
   1 mL of 0.002 M sodium thiosulfate is equivalent to 34 µg of oxidising substances, calculated as hydrogen peroxide.

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			Supersedes	Nil
Quality Control Department	Raw Material Standard Test Procedure		Effective Date	12/12/2020
	Item Name	Maize starch BP		
	Reference	BP	Market	Export

8. Iron :

Instruments : Analytical Balance.

Reagents : Hydrochloric acid, 20% w/v solution of iron free citric acid, Thioglycollic acid, Iron standard solution.

Procedure : Shake 1.5 g with 15 ml of dilute hydrochloric acid, Filter. The filtrate complies with the test. Use 10 ml filtrate, Add 2 ml of a 200 g/L solution of citric acid monohydrate R and 0.1 ml of thioglycollic acid. Mix, make alkaline with ammonia and dilute to 20 ml with water. Prepare a standard in the same manner, using 10 ml of iron standard solution (1 ppm Fe). After 5 min, any pink colour in the test solution is not more intense than that in the standard.

9. Loss On Drying:

Instruments : Analytical Balance, Oven.

Reagents : NA

Procedure: Weigh a glass stoppered weighing bottle (Wl) with as it is previously dried for 30 min at 130°±2°C. Transfer about 1.0 g of sample in bottle, cover it and accurately weigh (W2) distribute the sample as evenly as practicable by gently side wise shaking to a depth not exceeding 10mm. remove the stopper and dry the sample in an oven at 130° ±2°C for 90 minutes. Open the chamber. Close the bottle promptly and allow to cool in a desiccator before weighing. Weigh the bottle and record the weight (W3).

$$\text{Loss on drying (\%w/w)} = \frac{(W2-W3)}{(W2-W1)} \times 100$$

Where,

W_1= Weight of empty bottle

W_2 = Weight of bottle with sample

W_3 = Weight of bottle with sample after drying

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			Supersedes	Nil
Quality Control Department	Raw Material Standard Test Procedure		Effective Date	12/12/2020
	Item Name	Maize starch BP		
	Reference	BP	Market	Export

10. Sulphated Ash:

Instruments : Muffle Furnace, Analytical Balance.

Reagents : sulphuric acid

Procedure: Heat a silica or platinum crucible to redness for 10 minutes, allow to cool in a desiccator and weigh.(W1) Transfer 1.0g of sample to the crucible weigh the crucible and the contents accurately.(W2) Ignite, gently at first, until the sample is thoroughly charred. Cool, moisten the residue with 1 ml of sulphuric acid, heat gently until the white fumes are no longer evolved and ignite at 600° ± 50°C until all black particles have disappeared. Conduct the ignition in a place protected from air currents. Allow the crucible to cool, add a few drops of sulphuric acid and heat. Ignite as before, allow to cool and weigh(W4). Repeat the operation until two successive weighing do not differ by more than 0.5 mg to get constant weight(W5)

$$\text{Sulphated Ash (\%w/w)} = \frac{(W5-W1)}{W3} \times 100$$

Where,

W1 = weight of empty crucible
W2 = weight of crucible with sample
W3 = (W1-W2)
W4 = weight after ignition
W5 = constant weight

11. Microbial Contamination :

Instruments : Analytical balance

Reagents : NA

Procedure : Refer Microbiology SOP Number : SOP/QCM/039

Remark: As per new format SOP No. SOP/QCD/086

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			Supersedes	Nil
Quality Control Department	Raw Material Standard Test Procedure		Effective Date	14/12/2020
	Item Name	Hypromellose BP		
	Reference	BP	Market	Export

1. **Description:**
   Instruments : Analytical Balance
   Reagents :NA
   Procedure: Take sufficient Test sample in a petri dish and inspect visually.

2. **Solubility :**
   Instruments : Analytical Balance
   Reagents : Acetone, Anhydrous Ethanol, Toluene.
   Procedure:
   Hot Water: Take 10 mg of sample in test tube add 100 ml of Hot water and check for solubility.
   Acetone: Take 10 g of sample in test tube add 100 ml of Acetone and check for solubility.
   Anhydrous Ethanol: Take 10 g of sample in test tube add 100 ml of Anhydrous Ethanol and check for solubility.
   Toluene: Take 10 g of sample in test tube add 100 ml of Toluene and check for solubility.
   Cold water: Take 10 g of sample in test tube add 100 ml of Cold water and check for solubility.

3. **Identification:**
   A. By Chemical:
   Instruments: Analytical Balance
   Reagents: NA
   Procedure: Take 1.0 g of sample and evenly distribute onto the surface of 100 ml of water R in a beaker, tapping the top of the beaker gently if necessary to ensure a uniform layer on the surface. Allow to stand for 1-2 minutes: the powder material aggregates on the surface.

   B. By Chemical:
   Instruments: Analytical Balance, Magnetic Stirrer, Viscometer.
   Reagents: NA

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			Supersedes	Nil
Quality Control Department	Raw Material Standard Test Procedure		Effective Date	14/12/2020
	Item Name	Hypromellose BP		
	Reference	BP	Market	Export

Procedure : Take 1.0 g of sample and evenly distribute into 100 ml of boiling water R, and stir the mixture using a magnetic stirrer with a bar 25 mm long: a slurry is formed and the particle do not dissolve. Allow the slurry to cool to 10 °C and stir using a magnetic stirrer: clear or slightly turbid solution occurs with its thickness dependent on the viscosity grade.

C. By Chemical

Instruments: Analytical Balance, Water bath

Reagents: NA

Procedure: Use 0.1 ml of the solution obtained in identification test B add 9ml of 90% V/ solution of sulfuric acid R, shake ,heat on a water-bath for exactly 3 minutes, immediately cool in an ice-bath, carefully add 0.6 ml of a 20g/L solution of Nihydrin R, shake and allow to stand at 25°C:red colour develops at first and changes to purple within a 100 minutes.

D. By Physical:

Instruments: Analytical Balance

Reagents: NA

Procedure: Use 2-3 ml of identification test B solution and place onto a glass slide as a thin film and allow the water to evaporate: a coherent clear film forms on the glass slide.

E. By Chemical:

Instruments: Analytical Balance, thermometer.

Reagents: NA

Procedure: Use 50.ml of the solution obtained in identification test B to 50.0 ml of water R in a beaker. Insert a thermometer into solution. Stir the solution on a magnetic stirrer/hot plate and begin heating. Increasing the temperature at a rate of 2-5°C per min. Determine the temperature at which turbidity increase begins to occur and designate the temperature as the flocculation temperature: the flocculation temperature is higher than the 50 °C

4. **Apperance of solution:**

Instruments : Analytical Balance, pH meter.

Reagents : hydrazine sulfate solution.

Procedure: For Clarity of the Solution:

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Solution S (Test solution): Weigh and transfer 1.25 g of test sample into 25 ml of volumetric flask add 10 ml of water, sonicate to dissolve and make up the volume with same solvent.

Standard of Opalescence: Dilute 15.0 ml of the primary opalescent suspension to 1000.0 ml with water. This suspension is freshly prepared and may be stored for up to 24h.

Primary opalescent suspension (formazin suspension): Dissolve 2.5 g of hexamethylenetetramine in 100 ml of ground glass stoppered flask, add 25.0 ml water. Add 25.0 ml of hydrazine sulfate solution. Mix and allow to stand for 24h. This suspension is stable for 2 months, provided it is stored in a glass container free from surface defects. The suspension must not adhere to the glass and must be mixed thoroughly before use.

Hydrazine Sulfate Solution: Dissolve 1.0 g of hydrazine sulfate in water and dilute to 100.0 ml with same solvent. Allow to stand for 4-6h.

Reference Suspensions:

Prepare the reference suspensions given below and compare with test solution for clarity of solution.

	I	II	III	IV
Standard of opalescence	5.0 ml	10.0 ml	30.0 ml	50.0 ml
Water	95.0 ml	90.0 ml	70.0 ml	50.0 ml

A liquid is considered clear if its clarity is the same as that of water or of the solvent used for preparing the solution under examination or if its opalescence is not more than that of opalescence standard OS1.

For Colour of the Solution:

Using identical test tubes of colourless, transparent, neutral glass with a flat base and an internal diameter of 15 – 25 mm, compare the liquid to be examined with a reference suspension freshly prepared as described below. Ensure that the depths of the layers in the 2 test-tubes are the same (about 40 mm)

Standard Solutions

Standard Solution	Volumes in ml			
	Yellow Solution	Red Solution	Blue Solution	HCl (10g/L)
B (Brown)	3.0	3.0	2.4	1.6
BY (Brownish-Yellow)	2.4	1.0	0.4	6.2

Y (yellow)	2.4	0.6	0.0	7.0
GY (Greenish-Yellow)	9.6	0.2	0.2	0.0
R (red)	1.0	2.0	0.0	7.0

For Reference Solution Y

Reference Solution	Volumes in ml	
	Standard Solution Y	HCl (10g/L)
Y1	100.0	0.0
Y2	75.0	25.0
Y3	50.0	50.0
Y4	25.0	75.0
Y5	12.5	87.5
Y6	5.0	95.0
Y7	2.5	97.5

A solution is colourless if it has appearance of water or solvent used for preparation of solution to be examined, or is not more intensely coloured than reference solution.

5. **pH:**

Instruments : Analytical Balance, pH meter.

Reagents : NA

Procedure: Dissolve 1 g in water R and dilute to 10 mL with the same solvent. And note down the pH by using calibrated pH meter.

6. **Viscosity:**

Instruments: Analytical Balance, Viscometer.

Reagents : NA

Procedure:

Method 1: Take 4.000 g of dried sample. Transfer into a wide mouth bottle, and add 200.0 g hot water and capping the bottle, stir by mechanical means at 400 ± 50 r/min for 10-20 min until the particles are thoroughly disperse and wetted. Continue the stirring in cooling water bath maintained at temperature below 10°C for another 20-40 min, determine the kinematic

viscosity (v) of this solution using capillary viscometer method. separately determine the density (ρ) of the solution and calculate the dynamic viscosity (η), as η=ρv.

7. **Loss on Drying:**

Instruments : Analytical Balance, Oven

Reagents : NA

Procedure: Weigh a glass stoppered weighing bottle (Wl) with as it is previously dried for 30 min at 300 ± 15 °C. Transfer about 1 g of sample in bottle, cover it and accurately weigh (W2) distribute the sample as evenly as practicable by gently side wise shaking to a depth not exceeding 10mm. remove the stopper and dry the sample in an oven at 300 ± 15 °C. For 1 hr. Open the chamber close the bottle promptly and allow to in a desiccator before weighing, weigh the bottle and record the weight (W3).

$$\text{Loss on drying (\%w/w)} = \frac{(W2-W3)}{(W2-W1)} \times 100$$

Where,

W_1 = Weight of empty bottle

W_2 = Weight of bottle with sample

W_3 = Weight of bottle with sample after drying

8. **Sulfated ash:**

Instruments : Muffle Furnace, Analytical Balance.

Reagents : sulphuric acid

Procedure: Heat a silica or platinum crucible to redness for 10 minutes, allow to cool in a desiccator and weigh. (W1) Transfer 1g of sample to the crucible weigh the crucible and the contents accurately.(W2) Ignite, gently at first, until the sample is thoroughly charred. Cool, moisten the residue with 1 ml of sulphuric acid, heat gently until the white fumes are no longer evolved and ignite at 600° ± 50° until all black particles have disappeared. Conduct the ignition in a place protected from air currents. Allow the crucible to cool, add a few drops of sulphuric acid and heat. Ignite as before, allow to cool and weigh (W4). Repeat the operation until two successive weighing do not differ by more than 0.5 mg to get constant weight (W5).

$$\text{Sulphated Ash (\%w/w)} = \frac{(W5-W1)}{W3} \times 100$$

Where,
W1 = weight of empty crucible
W2 = weight of crucible with sample
W3 = (W1-W2)
W4 = weight after ignition
W5 = constant weight

9. Assay:

Instruments :Analytical Balance, GC

Reagents : adipic acid, hydriodic acid, isopropyl iodide, methyl iodide.

Procedure:

Test solution: Weigh 65.0 mg of substance to examined, place in a reaction vial, add 0.06-0.10 g of adipic acid R, 2.0 ml of the internal standard solution and 2.0 ml of hydriodic acid R, immediately cap and seal the vial, and weigh accurately. Mix the contents of the vial continuously for the 60 min. while heating the block so that the temperature of the contents is maintained at 130 ± 2°C. If a reciprocal shaker or magnetic stirrer cannot be used, shake the vial thoroughly by hand at 5 min intervals during the initial 30 min of the heating time. Allow the vial to cool, and again weigh accurately. If the loss of mass is less than 26 mg and there is no evidence of a leak, use the upper layer of the mixture as the test solution.

Reference solution: Take 0.06-0.10 g of Adipic acid R, 2.0 ml of the internal standard solution and 2.0 ml of Hydriodic acid R in another reaction vial, cap and seal the vial, and weigh accurately. Add 15-22 µL of isopropyl iodide R through the septum with a syringe, weigh accurately, add 45 µL of methyl iodide R in the same manner, and weigh accurately. Shake the reaction vial thoroughly and use the upper layer as the reference solution.

Column:

— material: fused silica;
— size: l = 30 m, Ø = 0.53 mm;
— stationary phase: Poly(dimethyl siloxane)R (3 µm).
Carrier gas helium for chromatography R.

Flow rate 4.3 mL/min
Split ratio 1:40.

Temperature:

	Time (min)	Temperature (°C)
Column	0-3	50
	3-8	50 → 100
	8-12.3	100 → 250
	12.3-20.3	250
Injection port		250
Detector		280

Detection Flame ionisation; make-up gas flow rate: 25 mL/min.
Injection 1-2 µL.
Relative retention :With reference to octane (retention time = about 10 min): methyl iodide = about 0.4; isopropyl iodide = about 0.7.
System suitability: reference solution (b):
— resolution: minimum 5.0 between the peaks due to methyl iodide and isopropyl iodide and between the peaks due to isopropyl iodide and octane:
— repeatability: maximum relative standard deviation of 2.0 per cent for the ratios of the areas of the peaks respectively due to methyl iodide and isopropyl iodide to the area of the peak due to octane, determine on 6 injections.
calculate the ratios (Q1 and Q2) of the areas of the peaks due to methyl iodide and isopropyl iodide to the area of the peak due to the internal standard in the chromatogram obtained with the test solution, and the ratios (Q1 and Q2) of the areas of the peaks due to methyl iodide and isopropyl iodide of the area of the peak due to the internal standard in the chromatogram obtain with reference solution.

Calculation of percentage content of Methoxy groups using the following expression:

$$\frac{Q1}{Q3} \times \frac{m1}{m} \times 21.864$$

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			Supersedes	Nil
Quality Control Department	Raw Material Standard Test Procedure		Effective Date	14/12/2020
	Item Name	Hypromellose BP		
	Reference	BP	Market	Export

Calculation of percentage content of Hydroxypropoxy groups using the following expression:

$$\frac{Q2}{Q4} \times \frac{m2}{m} \times 44.17$$

$m1$ = mass of the methyl iodide in the reference solution, in milligrams;
$m2$= mass of isopropyl iodide in the reference solution, in milligrams;
m =mass of the sample dried substance, in milligrams.

Remark: As per new format SOP No. SOP/QCD/086

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			Supersedes	Nil
Quality Control Department	Raw Material Standard Test Procedure		Effective Date	15/12/2020
	Item Name	Magnesium Stearate BP		
	Reference	BP	Market	Export

1. **Description:**
   Instruments :Analytical Balance.
   Reagents :NA
   Procedure: Take sufficient Test sample in a petri dish and inspect visually.

2. **Solubility :**
   Instruments :Analytical Balance.
   Reagents :NA
   Procedure: Practically insoluble in water and in anhydrous ethanol.

3. **Identification:**
   A. Freezing Point:
   Instruments : Analytical Balance.
   Reagents : NA
   Procedure :
   Solution S: To 5.0 g add 50 ml of peroxide-free ether, 20 ml of dilute nitric acid and 20 ml of water and heat under a reflux condenser until dissolution is complete. Allow to cool. In a separating funnel, separate the aqueous layer and shake the ether layer with 2 quantities, each of 4 ml, of water . Combine the aqueous layers, wash with 15 ml of peroxide-free ether and dilute to 50.0 mL with water (solution S). Evaporate the organic layer to dryness and dry the residue at 100-105 °C. Keep the residue for identification tests A and B.

   Determined on the residue obtained in the preparation of solution S.
   Place in the inner tube sufficient quantity of the liquid or previously melted substance to be examined, to cover the thermometer bulb and determine the approximate freezing point by cooling rapidly. Place the inner tube in a bath about 5 °C above the approximate freezing point until all but the last traces of crystals are melted. Fill the beaker with water or a saturated solution of sodium chloride, at a temperature about 5 °C lower than the expected freezing point, insert the inner tube into the outer tube, ensuring that some seed crystals are present,

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and stir thoroughly until solidification takes place. Note the highest temperature observed during solidification.

B. Acid value:

Instruments : Analytical Balance.

Reagents : Ethanol, Ether, 0.1 M Potassium hydroxide, Phenolphthalein.

Procedure : Dissolve 10.00 g of the substance in 50 ml of a mixture of equal volumes of ethanol (96 per cent) and light petroleum, previously neutralized with 0.1 M potassium hydroxide or 0.1 M sodium hydroxide, unless otherwise specified, using 0.5 ml of phenolphthalein solution as indicator. If necessary, heat to about 90 °C to dissolve the substance to be examined. When the substance to be examined has dissolved, titrate with 0.1 M potassium hydroxide or 0.1 M sodium hydroxide until the pink colour persists for at least 15 s (n ml of titrant). When heating has been applied to aid dissolution, maintain the temperature at about 90 °C during the titration.

Acid Value = 5.611 × n/w

Where n= the number of ml of 0.1 M Potassium hydroxide required

w= weight in g of the substance

C. By GC:

Instruments : NA

Reagents : NA

Procedure : The 2 principal peaks in the chromatogram obtained with the test solution are similar in retention time to the 2 principal peaks in the chromatogram obtained with the reference solution.

D. By Chemical Reaction:

Instruments : NA

Reagents : Ammonium chloride solution, dilute ammonia, disodium hydrogen phosphate dodecahydrate.

Procedure : To 1 ml of solution S add 1 ml of dilute ammonia; a white precipitate is formed that dissolves on addition of 1 ml of ammonium chloride solution. Add 1 ml of a 120 g/L solution of disodium hydrogen phosphate dodecahydrate; a white crystalline precipitate is formed.

4. **Acidity or Alkalinity:**

Instruments : Analytical Balance.

Reagents : Bromothymol Blue solution, 0.1M Hydrochloric acid.

Procedure : To 1.0 g add 20 ml of carbon dioxide-free water and boil for 1 min with continuous shaking. Cool and filter. To 10 ml of the filtrate add 0.05 ml of bromothymol blue solution. Not more than 0.05 ml of 0.1 M hydrochloric acid or 0.1 M sodium hydroxide is required to change the colour of the indicator.

5. **Chlorides:**

Instruments : Analytical Balance.

Reagents : Nitric acid, Hydrochloric acid

Procedure : Dilute 10.0 ml of solution S to 40 ml with water. Neutralize if necessary with nitric acid using litmus as indicator. Add 1 ml of nitric acid and 1 ml of 0.1 M silver nitrate and dilute to 50 ml with water. Mix and allow to stand for 5 min protected from light. The turbidity, if any, is not greater than that produced in a solution containing 1.4 ml of 0.02 M hydrochloric acid.

6. **Sulphates:**

Instruments : Analytical Balance.

Reagents : Hydrochloric acid, barium chloride solution, sulfuric acid.

Procedure : Dilute 6.0 ml of solution to 40 mL with water. Neutralize if necessary with hydrochloric acid using litmus as indicator. Add 1 ml of 3 M hydrochloric acid and 3 ml of a 120 g/L solution of barium chloride Rand dilute to 50 ml with water. Mix and allow to stand for 10 min. The turbidity, if any, is not greater than that produced in a solution containing 3.0 mL of 0.02 M sulfuric acid.

7. **Cadmium: By Atomic Absorption spectrometry**

Instruments : Analytical balance, AAS

Reagents : Nitric acid, Ammonium dihydrogen phosphate

For the preparation of all aqueous solutions and for the rinsing of glassware before use, employ water that has been passed through a strong-acid, strong-base, mixed-bed ion-exchange resin before use. Select all reagents to have as low a content of cadmium, lead and nickel as practicable and store all reagent solutions in containers of borosilicate glass. Clean glassware before use by soaking in warm 8 M nitric acid for 30 min and by rinsing with deionised water.

Blank solution: Dilute 25 mL of cadmium- and lead-free nitric acid R to 100.0 mL with water R.

Modifier solution Dissolve 20 g of ammonium dihydrogen phosphate R and 1 g of magnesium nitrate R in water R and dilute to 100 mL with the same solvent. Alternatively, use an appropriate matrix modifier as recommended by the graphite furnace atomic absorption (GFAA) spectrometer manufacturer.

Test solution: Place 0.100 g of the substance to be examined in a polytetrafluoroethylene digestion bomb and add 2.5 mL of cadmium- and lead-free nitric acid R. Close and seal the bomb according to the manufacturer's operating instructions (when using a digestion bomb, be thoroughly familiar with the safety and operating instructions. Carefully follow the bomb manufacturer's instructions regarding care and maintenance of these digestion bombs. Do not use metal jacketed bombs or liners which have been used withhydrochloric acid due to contamination from corrosion of the metal jacket by hydrochloric acid). Heat the bomb in an oven at 170 °C for 3 h. Cool the bomb slowly in air to room temperature according to the bomb manufacturer's instructions. Place the bomb in a fume cupboard and open carefully as corrosive gases may be expelled. Dissolve the residue in water R and dilute to 10.0 mL with the same solvent.

Reference solution: Prepare a solution of 0.0030 µg/mL of Cd by suitable dilutions of a 0.00825 µg/mL solution of cadmium nitrate tetrahydrate R in the blank solution.

Dilute 1.0 mL of the test solution to 10.0 mL with the blank solution. Prepare mixtures of this solution, the reference solution and the blank solution in the following proportions: (1.0:0:1.0 V/V/V), (1.0:0.5:0.5 V/V/V), (1.0:1.0:0 V/V/V). To each mixture add 50 µL of modifier solution and mix. These solutions contain respectively 0 µg, 0.00075 µg and 0.0015 µg of cadmium per millilitre from the reference solution (keep the remaining test solution for use in the test for lead and nickel).

Source Cadmium hollow-cathode lamp.
Wavelength - 228.8 nm.
Atomisation device - Furnace.
Platform Pyrolytically coated with integrated tube.
Operating conditions Use the temperature programme recommended for cadmium by the GFAA spectrometer manufacturer. An example of temperature parameters for GFAA analysis of cadmium is shown below.

8. Lead: By Atomic Absorption spectrometry

Instruments : Analytical balance, AAS
Reagents : 8M nitric acid, lead standard solution
Procedure : For the preparation of all aqueous solutions and for the rinsing of glassware before use, employ water that has been passed through a strong-acid, strong-base, mixed-bed ion-exchange resin before use. Select all reagents to have as low a content of cadmium, lead and nickel as practicable and store all reagent solutions in containers of borosilicate glass. Clean glassware before use by soaking in warm 8 M nitric acid for 30 min and by rinsing with deionised water.
Blank solution: Use the solution described in the test for cadmium.
Modifier solution Use the solution described in the test for cadmium.
Test solution: Use the solution described in the test for cadmium.
Reference solution: Prepare a solution of 0.100 µg/mL of Pb by suitable dilutions of lead standard solution (100 ppm Pb) R with the blank solution.
Prepare mixtures of the test solution, the reference solution and the blank solution in the following proportions: (1.0:0:1.0 V/V/V), (1.0:0.5:0.5 V/V/V), (1.0:1.0:0 V/V/V). To each mixture add 50 µL of modifier solution and mix. These solutions contain respectively 0 µg, 0.025 µg and 0.05 µg of lead per millilitre from the reference solution.
Source Lead hollow-cathode lamp.
Wavelength 283.3 nm.
Atomisation device Furnace.
Platform Pyrolytically coated with integrated tube.

9. Nickel: By Atomic Absorption spectrometry

Instruments : NA

Reagents : 8M nitric acid, Nickel nitrate hexahydrate.

Procedure :

For the preparation of all aqueous solutions and for the rinsing of glassware before use, employ water that has been passed through a strong-acid, strong-base, mixed-bed ion-exchange resin before use. Select all reagents to have as low a content of cadmium, lead and nickel as practicable and store all reagent solutions in containers of borosilicate glass. Clean glassware before use by soaking in warm 8 M nitric acid for 30 min and by rinsing with deionised water.

Blank solution: Use the solution described in the test for cadmium.

Modifier solution Dissolve 20 g of ammonium dihydrogen phosphate R in water R and dilute to 100 mL with the same solvent. Alternatively, use an appropriate matrix modifier as recommended by the GFAA spectrometer manufacturer.

Test solution: Use the solution described in the test for cadmium.

Reference solution: Prepare a solution of 0.050 µg/mL of Ni by suitable dilutions of a 0.2477 µg/mL solution of nickel nitrate hexahydrate R in the blank solution.

Prepare mixtures of the test solution, the reference solution and the blank solution in the following proportions: (1.0:0:1.0 V/V/V), (1.0:0.5:0.5 V/V/V), (1.0:1.0:0 V/V/V). To each mixture add 50 µL of matrix modifier solution and mix. These reference solutions contain respectively 0 µg, 0.0125 µg and 0.025 µg of nickel per millilitre from the reference solution.

Source Nickel hollow-cathode lamp.

Wavelength 232.0 nm.

Atomisation device Furnace.

Platform Pyrolytically coated with integrated tube.

10. Loss On Drying:

Instruments : Analytical Balance, Oven.

Reagents : NA

Procedure: Weigh a glass stoppered weighing bottle (Wl) with as it is previously dried for 30 min at 105°±2°C. Transfer about 1.0 g of sample in bottle, cover it and accurately weigh (W2) distribute the sample as evenly as practicable by gently side wise shaking to a

depth not exceeding 10mm. remove the stopper and dry the sample in an oven at 105° ±2°C for 1 hours. Open the chamber. Close the bottle promptly and allow to cool in a desiccator before weighing. Weigh the bottle and record the weight (W3).

$$\text{Loss on drying (\%w/w)} = \frac{(W2-W3)}{(W2-W1)} \times 100$$

Where,

W_1= Weight of empty bottle

W_2 = Weight of bottle with sample

W_3 = Weight of bottle with sample after drying

11. Microbial Contamination: Refer Microbiology SOP Number : SOP/QCM/039

12. Assay

A. Magnesium:

Instruments : NA

Reagents : Ammonia solution, Ammonia buffer pH 10.0, 1M disodium edetate, 1 butanol, ethanol, mordant black II, 0.1M Zinc sulphate

Preparation of Reagent:

Ammonia buffer pH 10.0 (Ammonium chloride buffer pH 10.0): Dissolve 5.4 g of ammonium chloride in 20 ml of water, add 35 ml of 10 M ammonia and dilute with water to 100 ml.

Mordant black II mixture (Eriochrome Black T solution): Dissolve 0.2 g of eriochrome Black T and 2 g of hydroxylamine hydrochloride in sufficient methanol to produce 50 ml. (Note: prepare immediately before use).

Procedure: To 0.500 g in a 250 ml conical flask add 50 ml of a mixture of equal volumes of anhydrous ethanol and butanol, 5 ml of concentrated ammonia, 3 ml of ammonium

chloride buffer solution pH 10.0 , 30.0 ml of 0.1 M sodium edetate and 15 mg of mordant black 11 triturate. Heat at 45-50 °C until the solution is clear and titrates with 0.1 M zinc sulfate until the colour changes from blue to violet. Carry out a blank titration.

1 ml of 0.1 M sodium edetate is equivalent to 2.431 mg of Mg.

Formula:

$$\frac{\text{Diff. B.R. X Factor X Actual molarity X 100}}{\text{wt. of sample in mg X Standard molarity}}$$

B. Fatty acid Composition: Determine By Gas Chromatography

Test Solution Preparation: Dissolve 100 mg of sample in 5 ml of boron trifluoride solution. Boil under a reflux condenser for 10 min , add 4 ml of heptanes through the condenser and add 20.0 ml of a sodium chloride solution . Shake and allow the layers to separate. Dry the organic layer over 0.1 g of anhydrous sodium sulphate. Dilute 1.0 ml of this solution to 10.0 ml with heptane.

Reference Solution: Dissolve 50.0 mg each of palmitic acid RS and stearic acid RS in 5.0 ml of boron trifluoride solution. Boil under a reflux condenser for 10 min , add 4 ml of heptanes through the condenser and add 20.0 ml of a sodium chloride solution. Shake and allow the layers to separate. Dry the organic layer over 0.1 g of anhydrous sodium sulphate. Dilute 1.0 ml of this solution to 10.0 ml with heptane.

Chromatographic System:

Column: A capillary column 30 m × 0.32 mm, packed with fused silica coated with macrogol 20000 (film thickness 0.5 um)

Temperature:

LEBEN LIFE SCIENCES	Leben Life Sciences Pvt. Ltd. Akola - 444104 (MS) INDIA		STP No.	LLS/STP/RM/023/00
			Supersedes	Nil
Quality Control Department	Raw Material Standard Test Procedure		Effective Date	15/12/2020
	Item Name	Magnesium Stearate BP		
	Reference	BP	Market	Export

Time (Min)	Temperature (°C)
0-2	70
2-36	70-240
36-41	240

- Inlet port at 220 °C and detector at 260 °C
- Flame Ionization Detector
- Flow rate: 2.4 ml per min, helium or nitrogen as the carrier gas.

Inject 1 ul of reference solution. The relative retention with reference to methyl stearate for methyl palmitate is about 0.9. The test is not valid unless the resolution between the peaks due to methyl stearate and methyl palmitate is not less than 5.0.

Inject 1 ul of reference solution and the test solution.

Calculate the percent content of stearic acid and palmitic acid.

Remark: As per new format SOP No. SOP/QCD/086

	Prepared By	Reviewed By	Approved By
Sign and Date	15/12/2020	15/12/2020	15/12/2020

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LEBEN LIFE SCIENCES	Leben Life Sciences Pvt. Ltd. Akola - 444104 (MS) INDIA	STP No.	LLS/STP/RM/024/00
		Supersedes	Nil
Quality Control Department	Raw Material Standard Test Procedure	Effective Date	16/12/2020
	Item Name: Readycoat Iron Oxide Yellow 36FO193220		
	Reference: IH	Market	Export

1. **Description:**
   Instruments : Analytical Balance
   Reagents : NA
   Procedure: Take 2 gm of test sample in a petri dish and inspect visually.

2. **Identification:**
   Instruments : NA
   Reagents : NA
   Procedure:
   Test Solution: Take 153g Isopropyl alcohol and 102 g of Acetone in a conical flask. And slowly add accurate 30 g of sample with continue stirring on magnetic stirrer. Close the flask and continue stirring 30 min.
   Take 1 to 5 ml of this solution, pour it on a glass slide allow to evaporate the solvent and observe the film formation.

3. **Particle Size:**
   Instrument : Analytical Balance, Sieve shaker
   Reagents : NA
   Procedure: Weigh empty sieve with mesh size 100 then weigh and transfer about 100 g of solution prepared in the test of identification through the sieve with mesh size 100# ASTM. Wash the sieve with solvent and dry at 110° C than cool to room temp. and weigh.

$$\text{Pass through 100\# sieve} = 100 - \frac{W3-W1}{W2} \times 100$$

Where,

W_1 = Weight of empty sieve
W_2 = weight of powdered dispersion
W_3 = Weight of sieve after drying

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Department	Quality Control	Quality Control	Quality Assurance

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4. **Loss on drying:**
Instruments : Analytical Balance, Oven.
Reagents : NA
Procedure: Weigh a glass stoppered weighing bottle (Wl) with as it is previously dried for 30 min at 110° C. Transfer about 1 g of sample in bottle, cover it and accurately weigh (W2) distribute the sample as evenly as practicable by gently side wise shaking to a depth not exceeding 10mm. remove the stopper and dry the sample in an oven at 110° ±2°C for 2 hours. Open the chamber Close the bottle promptly and allow the sample in a desiccator with phosphorus pentoxide bed to come room temperature, weigh the bottle and record the weight (W3).

$$\text{Loss on drying (\%w/w)} = \frac{(W2-W3)}{(W2-W1)} \times 100$$

Where,

W_1 = Weight of empty bottle

W_2 = Weight of bottle with sample

W_3 = Weight of bottle with sample after drying

5. **Arsenic:**
Instruments : Analytical Balance
Reagents : Arsenic trioxide, sodium hydroxide, hydrochloric acid, 1 M potassium Iodide, 2M sodium hydroxide
Procedure:
Arsenic standard solution (3 ppm): Dissolve 0.33 g of arsenic trioxide in 5 ml of 2M sodium hydroxide and dilute to 250.0 ml with water. Dilute 1 volume of this solution to 100 volumes with water. Further dilute 3 ml to 10 ml with same solvent.

Test Solution: Transfer 1 g of sample into generator flask and dilute to 35 ml with water.
Take 100-m1 bottle or conical flask closed with a rubber or ground glass stopper through which passes a glass tube (about 20 cm x 5 mm). The lower part of the tube is drawn to an internal diameter of 1.0 mm, and 15 mm from its tip is a lateral orifice 2 to 3 mm in diameter. When the tube is in position in the stopper the lateral orifice should be at least 3 mm below the lower surface of the stopper. The upper end of the tube has a perfectly flat surface at right angles to the axis of the tube. A second glass tube of the same internal diameter and 30 mm

long, with a similar flat surface, is placed in contact with the first and is held in position by two spiral springs or clips. Into the lower tube insert 50 to 60 mg of lead acetate cotton, loosely packed, or a small plug of cotton and a rolled piece of lead acetate paper weighing 50 to 60 mg. Between the flat surfaces of the tubes place disc or a small square of mercuric chloride paper large enough to cover the orifice of the tube (15 mm x 15 mm).

Into the bottle or conical flask introduce the test solution. add 5 ml of 1 M potassium iodide and 10 g of zinc AsT. Immediately assemble the apparatus and immerse the flask in a water-bath at a temperature such that a uniform evolution of gas is maintained. After 40 minutes any stain produced on the mercuric chloride paper is not more intense than that obtained by treating in the same manner 3.0 ml of arsenic standard solution (3 ppm As) diluted to 50 ml with water.

6. **Tapped Density (g/ml):**

Instruments : Analytical Balance, tapped density apparatus

Reagents : NA

Procedure : Weigh about 20g sample and transfer into a 100 ml measuring cylinder carry out 300 taps and note down the volume occupied by the material. Calculate the Tapped density as follows:

$$\text{Tapped density (g/ml)} = \frac{\text{weight of the sample (g)}}{\text{Volume occupied (ml)}}$$

Remark: As per new format SOP No. SOP/QCD/086

LEBEN LIFE SCIENCES	Leben Life Sciences Pvt. Ltd. Akola - 444104 (MS) INDIA	STP No.	LLS/STP/WS/001/00
		Supersedes	Nil
Quality Control Department	Standard Test Procedure Water Sample	Effective Date	13/01/2021
	Item Name: Purified Water		
	Reference: IH/IP/BP/USP	Market	Export

1. **Description:**

Instrument : Analytical Balance

Reagent: NA

Procedure: Take the 100ml water in clean test tube and observe.

2. **pH:**

Instrument : pH meter

Reagent : Potassium Chloride

Procedure: Take about 100 ml of sample and add 0.3 ml of Saturated KCl Solution and read pH at 25 °C on Suitable Calibrated pH Meter.

3. **Sulphate:**

Instrument : Analytical Balance

Reagents : Potassium Sulphate, Barium Chloride, Acetic Acid

Preparation of Reagents:

Sulphate Standard Solution (1000 ppm SO_4) 0.181% w/v:

Weigh accurately 0.181g of potassium sulphate, transfer to a 100ml volumetric flask add 30% ethanol, dissolve and dilute to 100 ml with the same solvent.

Sulphate Standard Solution (10 ppm SO_4) R1:

Dilute 1 ml solution of 1000ppm SO_4 and dilute to 100 ml with ethanol (30%)

Sulphate Standard Solution (1000 ppm SO_4) R:

Weigh accurately 0.181g of potassium sulphate, transfer to a 100ml volumetric flask add distilled water, dissolve and dilute to 100 ml with the same solvent.

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Date	13/01/2021	13/01/2021	13/01/2021
Department	Quality Control	Quality Control	Quality Assurance

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Sulphate Standard Solution (250 ppm SO_4):

Dilute 25 ml of sulphate standard solution (1000ppm SO_4) R and dilute to 100 ml with distilled water immediately before use.

25% w/v solution of barium chloride solution:

Dissolve 25 g of barium chloride in distilled water and dilute to 100ml with the same solvent.

5M Acetic acid:

Take 28.5 ml of acetic acid to 100 ml volumetric flask previously containing about 50 ml of distilled water and dilute up to the mark with distilled water.

Procedure:

Add 1 ml of a 25% w/v solution of barium chloride to 1.5 ml of sulphate standard solution (10 ppm SO_4) R1, shake and allow to stand for 1 minute. Add 10 ml of the sample, 0.5 ml of 5M acetic acid and allow to stand for a minute.

Any opalescence produced is not more intense than that of a standard prepared in the same manner by using 10 ml of sulphate standard solution (250 ppm SO_4) in place of the sample.

4. **Chloride:**

Instrument : Analytical Balance

Reagents : Nitric Acid, Silver Nitrate, Sodium Chloride

Preparation of Reagents:

2M Nitric acid solution:

Take 126 ml of nitric acid (70% w/w) to 1000 ml volumetric flask previously containing about 200 ml of distilled water and dilute up to the mark with distilled water.

Silver nitrate solution 1.7%:

Dissolve 1.7 g of silver nitrate in sufficient water and dilute to 100 ml with same solvent (1.7% w/v). Store protected from light.

Chloride Standard Solution (500 ppm Cl):

Dissolve 0.0824 g sodium chloride in sufficient distilled water and dilute to 100 ml with the same solvent.

Chloride Standard Solution (250 ppm Cl):

Dilute 25 ml of the standard solution (500ppm Cl) to 50 volumes with distilled water immediately before use.

Procedure:

To 10 ml of the sample add 1 ml of 2M nitric acid, pour the mixture as a single addition into 1 ml of silver nitrate solution and allow to stand for 5 minutes protected from light. When viewed transversely against a black background any opalescence produced is not more intense than that obtained by treating 10 ml of chloride standard solution (250 ppm Cl) in the same manner.

5. **Ammonium:**

Instrument : Analytical Balance

Reagents: Ammonium chloride, sodium hydroxide, potassium iodide, Mercury Iodide

Preparation of Reagents:

Ammonium Standard Solution (250 ppm NH4):

Dissolve 0.0741 g of ammonium chloride in 50 ml of distilled water and dilute to 100 ml with same solvent (0.0741% w/v).

Ammonium Standard Solution (2.5 ppm NH4):

Dilute 1 ml standard solution (250 ppm NH4) to 100 ml with distilled water.

Ammonium Standard Solution (1 ppm NH4):

Dilute 10ml standard solution 2.5ppm to 25 ml with water.

25% w/v Sodium Hydroxide Solution:

Dissolve 25 g of sodium hydroxide in 100 ml of distilled water.

Alkaline potassium tetraiodomercurate solution:

Dissolve 11 g of potassium iodide and 15 g of mercury (II) iodide in water and dilute to 100 ml with water. Immediately before use, mix the solution with an equal volume of a 25% w/v solution of sodium hydroxide.

Procedure: To 20 ml of the sample in a nessler's cylinder add 1 ml of alkaline potassium tetra-iodo mercurate solution, stopper the tube, mix and allow to stand for 5 min.

The substance to be examined complies with the test if any yellow colour produced in the test solution should not be more intense than in the standard solution.

6. **Nitrates:**

Instrument : Analytical Balance

Reagents : Diphenylamine, Sulphuric acid, Potassium permanganate, Barium hydroxide, Potassium chloride, Potassium nitrate

Preparation of Reagents:

Nitrate Standard Solution (100 ppm)

Dissolve 0.163 g of potassium nitrate to 100 ml with water. Dilute 1 volume of this solution to 10 volume with water.

Standard Solution (2 ppm)

Dilute 1 volume of nitrate standard solution (100 ppm.), to 50 volumes with water.

Diphenylamine Solution:

Dissolve 0.1 g of diphenylamine in 100 ml Sulphuric acid.

Nitrate-free water:

To 100 ml ofwater add about 5 mg each of potassium permanganate and barium hydroxide and distil using a distillation flask with a straight tube condenser fitted to its side arm and an adaptor attached to the end of the condenser.

Test Procedure:

Place 5 ml of sample in a test-tube immersed in iced water, add 0.4 ml of a 100 g/l solution of potassium chloride, 0.1 ml of diphenylamine solution and, drop wise with shaking, 5 ml

of nitrogen-free sulphuric acid. Transfer the tube to a water-bath at 50 °C. After 15 min, any blue colour in the solution is not more intense than that in a reference solution prepared at the same time in the same manner using a mixture of 4.5 ml of nitrate-free water and 0.5 ml of nitrate standard solution (2 ppm NO_3).

7. **Heavy Metals:**

Instrument : Analytical Balance

Reagents : Lead nitrate, Ammonium acetate, Hydrochloric acid, Ammonia, Thioacetamide

Preparation of Reagents:

Lead Standard Solution (0.1% Pb):

Dissolve 400 mg of lead nitrate in sufficient distilled water to produce 250ml.

Lead Standard Solution (100 ppm Pb):

Dilute 1 ml of lead standard solution (0.1% Pb) to 10 ml with distilled water immediately before use.

Lead Standard Solution (10 ppm Pb):

Dilute 1 ml of lead standard solution (100 ppm Pb) to 10 ml with distilled water immediately before use.

Lead Standard Solution (1 ppm Pb):

Dilute 1 ml of lead standard solution (10 ppm Pb) to 10 ml with distilled water immediately before use.

Acetate Buffer pH 3.5:

Dissolve 25 g of ammonium acetate in 25 ml of water and add 38 ml of 7M hydrochloric acid. Adjust the pH to 3.5 with either 2M hydrochloric acid or 6M ammonia and dilute to 100 ml with water.

Thioacetamide solution:

Dissolve 0.4 g of thioacetamide to 10 ml with water.

Glycerol 85%:

Diluted 85 ml of glycerol to 100 ml with water.

Thioacetamide reagent:

Add 10 ml of a mixture of 15 mlof 1M sodium hydroxide, 5 mL of water and 20 of glycerol (85%) to 2 ml of thioacetamide solution, heat in a water bath for 20 seconds, cool and use immediately.

Test Solution:

Heat 200 ml of sample in a glass-evaporating dish on a water-bath until the volume is reduced to 20 ml. Use 12 ml of the concentrated solution.

Reference Solution:

A mixture of 10 ml of lead standard solution (1 ppm Pb) and 2ml of the test solution.

Blank solution:

A mixture of 10 ml water and 2 ml of test solution.

Test Procedure:

To each solution add 2 ml of buffer pH 3.5, mix, and add 1.2 ml of thioacetamide reagent. Mix immediately. Examine the solution after 2 minutes. This test is invalid if the reference solution does not show a slight brown colour compared to the blank solution.

The substance to be examined complies with the test if any brown colour in the test solution is not more intense than in the reference solution.

8. **Oxidisable Substances:**

Instrument : Analytical Balance

Reagents :Potassium permanganate, Sulphuric acid

Preparation of Reagents:

Dilute Sulphuric acid solution: Add 5.5 ml of sulphuric acid to 60 ml of water, allow to cool and add sufficient water to produce 100 ml.

0.02M Potassium permanganate:

Dissolve 3.2 g of potassium permanganate in 1000 ml of water, heat on a water bath for 1 hour, cool and filter through a sintered-glass filter.

Procedure:

To 100 ml sample add 10 ml of dilute sulphuric acid solution and 0.1 ml of 0.02M potassium permanganate solution and boil for 5 min. Upon observation the solution remains faintly pink.

TOC: Record the TOC from online TOC Analyzer.

9. **Residue On Evaporation:**

Instrument : Analytical balance

Reagents : NA

Weigh dry clean empty beaker or evaporating dish which is previously dried for 30 min at 100-105°C and transfer 100 ml sample. Evaporate on a water-bath and dry in an oven at 100-105°C to constant weight. Cool to room temperature and weigh beaker with residue. Difference between weight of beaker with residue and empty beaker is weight of residue.

Calculation:

Weight of empty beaker or evaporating dish = W1

Weight of beaker or evaporating dish + Residue = W2

Weight of Residue (W2-W1) = W3

$$\text{Residue on Evaporation} = \frac{W3\ (g) \times 100 \times 10000}{\text{Sample Quantity taken}}$$

10. * Conductivity:

Instrument : Conductivity Meter

Reagents : NA

Procedure: Take a reading of conductivityfrom water generation and purification system through EQ/ENG/PWD/CTT/001.

11. * TOC:

Instrument : TOC Analyzer

Reagents : NA

Procedure: Take a reading of TOCfrom water generation and purification system through TOC-301.

12. Calcium And Magnesium:

Instrument : Analytical Balance

Reagents : Ammonium chloride, Ammonia, Disodium Edetate

Preparation of Reagents:

Ammonia buffer pH 10.0:

Dissolve 5.4 g of ammonium chloride in 20 ml of water, add 35 ml of ammonia and dilute with water to 100 ml.

Preparation of 0.01 M Disodium EdetateSolutions:

Weight accurately about 3.75 g of Disodium Edetate, dissolve in sufficient water and dilute to 1000 ml with same solvent.

Mordant Black 11 Triturate:

Mix 1part of Mordant Black 11 with 99 parts of sodium chloride.

Procedure:

To 100 ml of sample, add 2 ml of ammonia buffer pH 10.0, 50-mg of mordant black II mixture and 0.5 ml of 0.01 M disodium Edetate.

A pure blue colour is produced.

13. Carbon Dioxide:

Instrument : Analytical Balance

Reagent : Calcium Hydroxide

Preparation of Reagents:

Calcium Hydroxide Solution:

Shake 10 g of calcium hydroxide repeatedly with 1000 ml of water and allow to stand until clear.

Procedure:

Add 25 ml Calcium Hydroxide TS in 25 ml of sample being examined.

The mixture should be clear.

14. Acidity Or Alkalinity:

Instrument : Analytical Balance

Reagent: bromothymol blue, methyl red

To 10 ml of freshly boiled and cooled in a borosilicate glass flask ,add 0.05 ml methyl red solutionthe resulting solution is not red. To 10 ml of freshly boiled and cooled in a borosilicate glass flask, add 0.1 ml bromothymol blue solution the resulting solution is not Blue.

15. Microbial limit test: Refer Microbiology SOP Number : SOP/QCM/039

* Conductivity and TOC shall be performed only in case of sampling point SP-304 (In-return line of Purified Water Loop)

Remark: As per new format SOP No. SOP/QCD/086